

```
import pandas as pd
```

```
df=pd.read_csv('/content/StudentPerformanceFactors.csv')
df
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring_Sessions
0	23	84	Low	High	No	7	73	Low	Yes	
1	19	64	Low	Medium	No	8	59	Low	Yes	
2	24	98	Medium	Medium	Yes	7	91	Medium	Yes	
3	29	89	Low	Medium	Yes	8	98	Medium	Yes	
4	19	92	Medium	Medium	Yes	6	65	Medium	Yes	
...	...	...	...	...	...	...	...	...	...	...
6602	25	69	High	Medium	No	7	76	Medium	Yes	
6603	23	76	High	Medium	No	8	81	Medium	Yes	
6604	20	90	Medium	Low	Yes	6	65	Low	Yes	
6605	10	86	High	High	Yes	6	91	High	Yes	
6606	15	67	Medium	Low	Yes	9	94	Medium	Yes	

6607 rows × 20 columns

```
df.head()
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring_Sessions
0	23	84	Low	High	No	7	73	Low	Yes	
1	19	64	Low	Medium	No	8	59	Low	Yes	
2	24	98	Medium	Medium	Yes	7	91	Medium	Yes	
3	29	89	Low	Medium	Yes	8	98	Medium	Yes	
4	19	92	Medium	Medium	Yes	6	65	Medium	Yes	

```
df.tail()
```

[illegible]

```
df.describe(include='all')
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring
count	6607.000000	6607.000000	6607	6607	6607	6607.000000	6607.000000	6607	6607	6607
unique	NaN	NaN	3	3	2	NaN	NaN	3	2	2
top	NaN	NaN	Medium	Medium	Yes	NaN	NaN	Medium	Yes	Yes
freq	NaN	NaN	3362	3319	3938	NaN	NaN	3351	6108	6108
mean	19.975329	79.977448	NaN	NaN	NaN	7.02906	75.070531	NaN	NaN	NaN
std	5.990594	11.547475	NaN	NaN	NaN	1.46812	14.399784	NaN	NaN	NaN
min	1.000000	60.000000	NaN	NaN	NaN	4.00000	50.000000	NaN	NaN	NaN
25%	16.000000	70.000000	NaN	NaN	NaN	6.00000	63.000000	NaN	NaN	NaN
50%	20.000000	80.000000	NaN	NaN	NaN	7.00000	75.000000	NaN	NaN	NaN
75%	24.000000	90.000000	NaN	NaN	NaN	8.00000	88.000000	NaN	NaN	NaN
max	44.000000	100.000000	NaN	NaN	NaN	10.00000	100.000000	NaN	NaN	NaN

```
df[['Hours_Studied', 'Attendance']].describe()
```

	Hours_Studied	Attendance
count	6607.000000	6607.000000
mean	19.975329	79.977448
std	5.990594	11.547475
min	1.000000	60.000000
25%	16.000000	70.000000
50%	20.000000	80.000000
75%	24.000000	90.000000
max	44.000000	100.000000

```
Scores=df.filter(like='Score')
Scores
```

	Previous_Scores	Exam_Score
0	73	67
1	59	61
2	91	74
3	98	71
4	65	70
...	...	...
6602	76	68
6603	81	69
6604	65	68
6605	91	68
6606	94	64

6607 rows × 2 columns

```
Scores=df.filter(like='Exam_Score')
Scores
```

	Exam_Score
0	67
1	61
2	74
3	71
4	70
...	...
6602	68
6603	69
6604	68
6605	68
6606	64

6607 rows × 1 columns

```
high_score=df[df['Exam_Score'] > 70]
high_score.head()
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring_Sessions
2	24	98	Medium	Medium	Yes	7	91	Medium	Yes	
3	29	89	Low	Medium	Yes	8	98	Medium	Yes	
5	19	88	Medium	Medium	Yes	8	89	Medium	Yes	
9	23	98	Medium	Medium	Yes	8	71	Medium	Yes	
11	17	97	Medium	High	Yes	6	87	Low	Yes	

```
df=pd.read_csv('/content/StudentPerformanceFactors.csv')
gender = df[['Gender']]
gender
```

Gender	
0	Male
1	Female
2	Male
3	Male
4	Female
...	...
6602	Female
6603	Female
6604	Female
6605	Female
6606	Male

6607 rows × 1 columns

```
male_students=df[df['Gender']=='Male']
female_students=df[df['Gender']=='Female']
```

```
male_students.head()
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring_Sessions
0	23	84	Low	High	No	7	73	Low	Yes	
2	24	98	Medium	Medium	Yes	7	91	Medium	Yes	
3	29	89	Low	Medium	Yes	8	98	Medium	Yes	
5	19	88	Medium	Medium	Yes	8	89	Medium	Yes	
6	29	84	Medium	Low	Yes	7	68	Low	Yes	

```
female_high_scorers = df[(df['Gender'] == 'Female') & (df['Exam_Score'] > 70)]
female_high_scorers.head()
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring
73	29	92	Low	Medium	No	4	91	Medium	Yes	
94	18	89	High	Medium	Yes	4	73	Medium	Yes	
113	35	99	High	High	Yes	7	85	Low	Yes	
115	22	89	High	High	Yes	6	80	Medium	Yes	
118	27	97	Low	High	Yes	8	81	Medium	Yes	

```
df.describe(include='all')
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring
count	6607.000000	6607.000000	6607	6607	6607	6607.00000	6607.000000	6607	6607	
unique	NaN	NaN	3	3	2	NaN	NaN	3	2	
top	NaN	NaN	Medium	Medium	Yes	NaN	NaN	Medium	Yes	
freq	NaN	NaN	3362	3319	3938	NaN	NaN	3351	6108	
mean	19.975329	79.977448	NaN	NaN	NaN	7.02906	75.070531	NaN	NaN	
std	5.990594	11.547475	NaN	NaN	NaN	1.46812	14.399784	NaN	NaN	
min	1.000000	60.000000	NaN	NaN	NaN	4.00000	50.000000	NaN	NaN	
25%	16.000000	70.000000	NaN	NaN	NaN	6.00000	63.000000	NaN	NaN	
50%	20.000000	80.000000	NaN	NaN	NaN	7.00000	75.000000	NaN	NaN	
75%	24.000000	90.000000	NaN	NaN	NaN	8.00000	88.000000	NaN	NaN	
max	44.000000	100.000000	NaN	NaN	NaN	10.00000	100.000000	NaN	NaN	

```
male_high_score=df[(df['Gender']=='Male') & (df['Exam_Score'] > 70)]
male_high_score.head()
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring
2	24	98	Medium	Medium	Yes	7	91	Medium	Yes	
3	29	89	Low	Medium	Yes	8	98	Medium	Yes	
5	19	88	Medium	Medium	Yes	8	89	Medium	Yes	
9	23	98	Medium	Medium	Yes	8	71	Medium	Yes	
11	17	97	Medium	High	Yes	6	87	Low	Yes	

```
female_high_scorers = df[(df['Gender'] == 'Female') & (df['Exam_Score'] > 70)]
female_high_scorers.head()
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring
73	29	92	Low	Medium	No	4	91	Medium	Yes	
94	18	89	High	Medium	Yes	4	73	Medium	Yes	
113	35	99	High	High	Yes	7	85	Low	Yes	
115	22	89	High	High	Yes	6	80	Medium	Yes	
118	27	97	Low	High	Yes	8	81	Medium	Yes	

```
count=df['Gender'].value_counts()
count
```

```
count
Gender
Male    3814
Female  2793
dtype: int64
```

```
import numpy as np
```

```
exam_score_array=df['Exam_Score'].to_numpy()
```

```
exam_score_array
```

```
array([67, 61, 74, ..., 68, 68, 64])
```

```
exam_score_array_2D=df['Exam_Score'].to_numpy().reshape(-1,1)
exam_score_array_2D
```

```
array([[67],
       [61],
       [74],
       ...,
       [68],
       [68],
       [64]])
```

```
mean=np.mean(exam_score_array_2D)
mean
```

```
np.float64(67.23565914938702)
```

```
max=np.max(exam_score_array_2D)
max
```

```
np.int64(101)
```

```
min=np.min(exam_score_array)  
min
```

```
np.int64(55)
```

```
median=np.median(exam_score_array)  
median
```

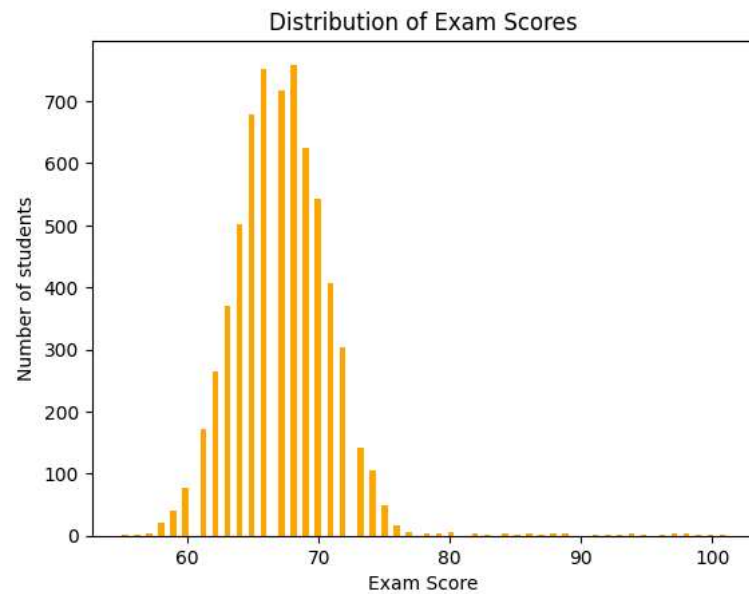
```
np.float64(67.0)
```

```
deviation=np.std(exam_score_array)  
deviation
```

```
np.float64(3.8901613508847057)
```

```
import matplotlib.pyplot as plt
```

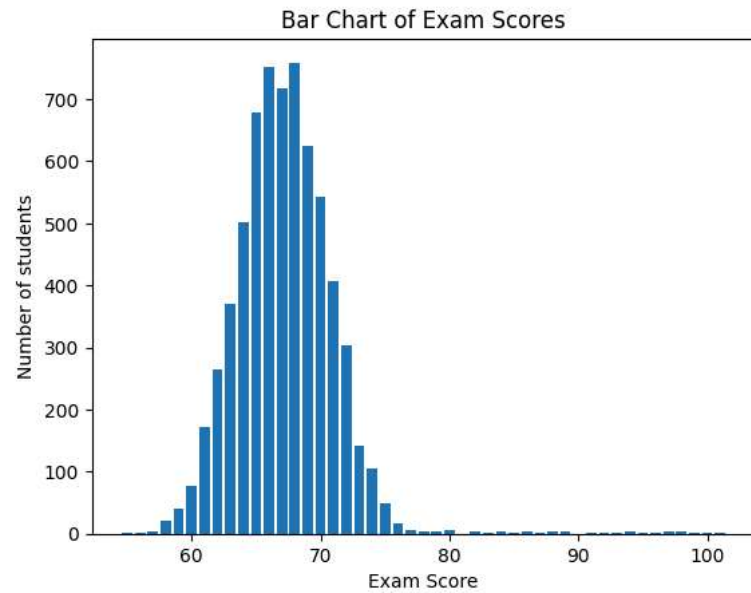
```
plt.hist(exam_score_array,color='orange',bins=100)  
plt.xlabel('Exam Score')  
plt.ylabel('Number of students')  
plt.title('Distribution of Exam Scores')  
plt.show()
```



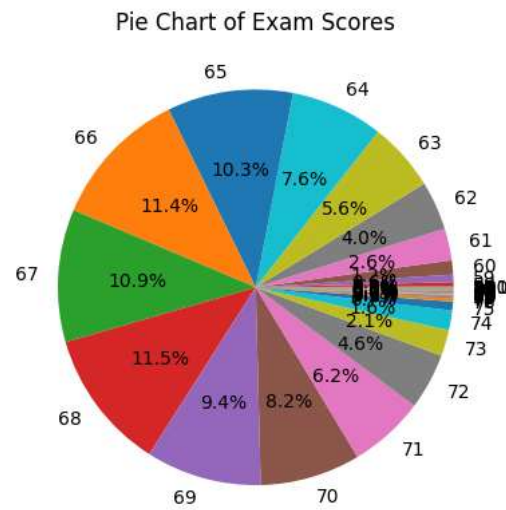
```
scores,counts=np.unique(exam_score_array,return_counts=True)  
plt.bar(scores,counts)  
plt.xlabel('Exam Score')  
plt.ylabel('Number of students')
```



```
plt.title('Bar Chart of Exam Scores')
plt.show()
```

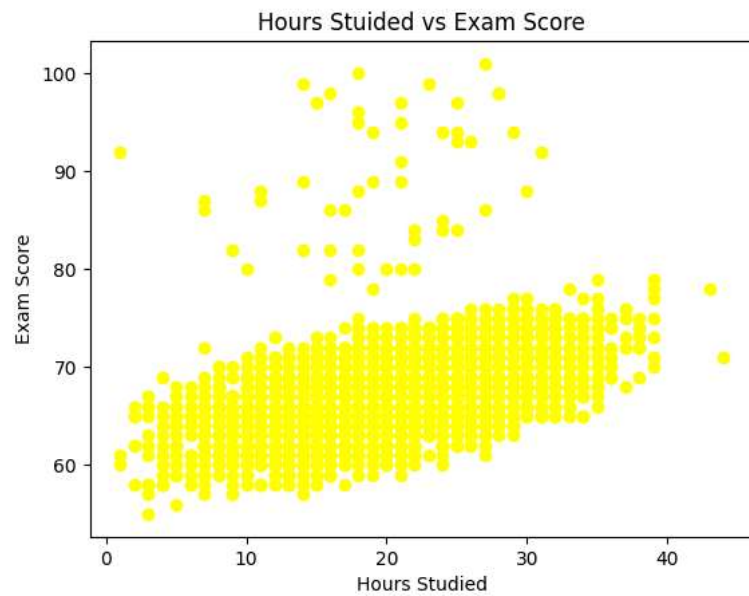


```
scores,counts=np.unique(exam_score_array,return_counts=True)
plt.pie(counts,labels=scores,autopct='%1.1f%%')
plt.title('Pie Chart of Exam Scores')
plt.show()
```



```
hourly_studied_array=df['Hours_Studied'].to_numpy()
```

```
plt.scatter(hourly_studied_array,exam_score_array,color='yellow')
plt.xlabel('Hours Studied')
plt.ylabel('Exam Score')
plt.title('Hours Stuided vs Exam Score')
plt.show()
```



Start coding or [generate](#) with AI.